

Exogenous Reservoir Volatility with an Exponential-Quadratic Readout

Current specification, physical stylised facts, and joint equity–volatility–index calibration architecture

Purpose. This document consolidates the current exogenous-reservoir candidate. It states the model under the physical and pricing measures, identifies the mechanisms supporting each targeted stylised fact, and separates mathematical implications from numerical evidence and open calibration questions.

Status: research specification, not a calibration result

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Abstract

The objective is a continuous-path stochastic-volatility model that can reproduce, under the physical measure, finite-frequency heavy tails, volatility clustering, roughness, leverage, a strong Zumbach effect and, at lower priority, the Taylor effect. Under the pricing measure, the same structural model should be capable of fitting Standard & Poor's 500 (S&P 500, whose option market is denoted SPX) and Cboe Volatility Index (VIX) option surfaces. The computational constraint is that changes in readout and risk-premium parameters must not require new simulation of the exogenous state paths.

The proposed model uses a structured bank of Ornstein–Uhlenbeck (OU) factors as a linear reservoir, an optional bounded nonlinear echo-state layer, and a normalised exponential-quadratic variance readout. A shifted quadratic projection of a return-correlated but variance-independent OU bank supplies leverage and time-reversal asymmetry. Slow exogenous OU modes supply clustering; a rough-kernel lift supplies finite-scale roughness; a fast, partly return-correlated direction supplies VIX-like spikes; and an independent quadratic block is reserved for dispersion or tail tuning. A deterministic Girsanov kernel preserves Gaussianity of the OU core between the physical probability measure $\mathbb{P}$ and the risk-neutral pricing measure $\mathbb{Q}$.

The architecture supports all requested mechanisms without a quintic polynomial. This is a capacity statement, not a proof that a single parameter vector fits every physical statistic and both option surfaces simultaneously. The main unresolved questions are empirical joint calibration, identifiability, and the placement of the upper tempering level. With bounded variance, true stock martingality follows on finite horizons by Novikov; without tempering, the exponential-quadratic moment and martingale domains must be enforced explicitly.

Contents

1	Objective and standard of claim	1
2	The current model	1
2.1	Driver system and measure convention	1
2.2	Layer 1: structured OU reservoir	2
2.3	Layer 2: optional bounded echo-state layer	2
2.4	Exponential-quadratic readout	3
2.5	Equity dynamics under $\mathbb{P}$ and $\mathbb{Q}$	4
2.6	Architecture at a glance	5
3	Parameter hierarchy and path reuse	5
3.1	Hyperparameters, structural parameters and daily inputs	5
3.2	What path reuse does and does not mean	6
4	Physical-measure stylised facts	7
4.1	Negligible linear autocorrelation of returns	7
4.2	Volatility clustering	7
4.3	Finite-scale roughness	8
4.4	Leverage effect	9

4.5	Strong Zumbach effect	10
4.6	Heavy-tailed returns at relevant frequencies	12
4.7	Taylor effect	13
4.8	Fast volatility, downside asymmetry and VIX spike relaxation	14
4.9	Summary of physical-measure support	16
5	Risk-neutral pricing and joint SPX–VIX calibration	16
5.1	Forward variance, VIX and VIX futures	16
5.2	Analytic Gaussian-core moments	16
5.3	Tempering and the nonlinear ESN layer	17
5.4	SPX pricing and martingality	17
5.5	Why a quintic polynomial is not structurally required	18
6	Joint calibration programme	18
6.1	Recommended nested sequence	18
6.2	Relation of M2 to ESN–A2	18
6.3	Calibration hierarchy	22
6.4	Validation and falsification tests	22
7	Open issues and limitations	23
8	Conclusion	24
A	Derivation of the Gaussian quadratic transform	24
B	Derivation of the Taylor benchmark	25
C	Related model classes	25

1 Objective and standard of claim

Let S denote the total-return-adjusted equity index level and V its instantaneous variance. Throughout, $\mathbb{P}$ denotes the physical probability measure governing historical dynamics, while $\mathbb{Q}$ denotes the risk-neutral probability measure used for derivative pricing. The programme has two linked objectives:

1. Under $\mathbb{P}$, reproduce the relevant empirical properties of equity returns and volatility at intraday and daily frequencies.
2. Under $\mathbb{Q}$, reproduce the cross section of SPX options, VIX futures and VIX options, while reusing the same exogenous random paths during calibration.

The phrases *supports* and *reproduces* must not be conflated. Four evidence levels are used throughout:

Label	Meaning	Interpretation
C	By construction	Follows directly from the state or readout definition.
P	Conditional proposition	Follows under explicit non-degeneracy, moment or scale assumptions.
N	Numerical mechanism evidence	Demonstrated by an ablation or controlled simulation, but not yet by market calibration.
O	Open empirical question	Must be established on historical data or option surfaces.

Main conclusion

There is no known structural incompatibility between the requested physical-measure facts, path reuse, and joint SPX–VIX pricing in the proposed class. There is also no theorem that a successful joint fit must exist. The correct conclusion is architectural feasibility subject to explicit moment, martingale and calibration tests.

2 The current model

2.1 Driver system and measure convention

Work on a filtered probability space satisfying the usual conditions. Under a measure $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$, let $W^{S,\mathbb{M}}$ be the standard Brownian motion driving the equity return. The superscript S labels the equity, and $\mathbb{M}$ identifies the probability measure. The structured reservoir uses four driver groups:

$$dW_t^{F,\mathbb{M}} = \rho_F dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_F^2} dB_t^{F,\mathbb{M}}, \quad (1)$$

$$dW_t^{J,\mathbb{M}} = \rho_J dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_J^2} dB_t^{J,\mathbb{M}}, \quad (2)$$

where $B^{F,\mathbb{M}}$ and $B^{J,\mathbb{M}}$ are independent standard Brownian motions, also independent of $W^{S,\mathbb{M}}$. The labels F and J denote the feedback and fast-spike banks, while ρ_F and ρ_J are their instantaneous correlations with the equity Brownian motion. Consequently $W^{F,\mathbb{M}}$ and $W^{J,\mathbb{M}}$

are standard Brownian motions correlated with the equity driver. The clustering drivers $B^{C,\mathbb{M}}$ and optional orthogonal drivers $B^{O,\mathbb{M}}$ are further independent standard Brownian motions, mutually independent and independent of W^S, B^F, B^J . Partial rather than zero correlation is recommended for the fast bank, because large VIX moves are empirically associated with equity sell-offs.

2.2 Layer 1: structured OU reservoir

For each bank $b \in \{F, J, C, O\}$, define under $\mathbb{P}$ the Ornstein–Uhlenbeck (OU) coordinates

$$dX_{j,t}^{b,\mathbb{P}} = -\lambda_j^b X_{j,t}^{b,\mathbb{P}} dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{P}}, \quad j = 1, \dots, N_b. \quad (3)$$

Here $X_{j,t}^{b,\mathbb{P}}$ is coordinate j of bank b at time t , $\lambda_j^b > 0$ is its mean-reversion rate, $W_j^{b,\mathbb{P}}$ is the Brownian motion assigned to that coordinate, and N_b is the number of coordinates in the bank. The diffusion coefficient $\sqrt{2\lambda_j^b}$ normalises every stationary marginal variance to one.

Within the feedback bank, the coordinates share W^F ; this creates the correlated multi-rate lift used to approximate a rough Volterra kernel. The fast bank can share one or a small number of partially return-correlated drivers. The clustering and orthogonal banks may use independent drivers, so that their covariance is a flexible positive mixture of exponentials.

For $N_b \geq 2$, the rate grids are geometric and fixed in the inner calibration:

$$\lambda_j^b = \lambda_{\min}^b \left(\frac{\lambda_{\max}^b}{\lambda_{\min}^b} \right)^{\frac{j-1}{N_b-1}}. \quad (4)$$

Thus $\lambda_{\min}^b$ and $\lambda_{\max}^b$ are respectively the slowest and fastest mean-reversion rates in bank b .

Define normalised projections

$$Y_t = a^\top X_t^F, \quad F_t = f^\top X_t^J, \quad \text{Var}_{\mathbb{P}}(Y_t) = \text{Var}_{\mathbb{P}}(F_t) = 1 \quad (5)$$

at the stationary physical reference point. Here X_t^F and X_t^J are the vectors of feedback- and fast-bank coordinates, while the fixed vectors a and f aggregate each bank into one scalar factor. The normalisation gives both projections unit stationary variance, so their readout coefficients have comparable interpretations.

The projection Y is a persistent, multiscale summary of past return-correlated shocks. Its shifted quadratic term below produces the main leverage effect—negative returns followed by higher volatility—and the Zumbach time-reversal effect. The projection F emphasises the fastest return-correlated modes. Its role is to create the sharp front of a volatility spike and its subsequent mean-reverting decay. Shifting the quadratic function of F makes a negative equity move generate a larger spike than a positive move of equal magnitude.

2.3 Layer 2: optional bounded echo-state layer

An echo-state network (ESN) is an optional second reservoir layer with fixed recurrent weights and a calibrated output loading. On a time grid $t_n = n\Delta t$, where n is the time-step index, $\Delta t > 0$ is the step length and $0 < \alpha \leq 1$ is the leak parameter, define

$$Z_{n+1} = (1 - \alpha)Z_n + \alpha \tanh(A_Z Z_n + B_Z \psi(X_n) + b_Z), \quad (6)$$

where Z_n is the second-layer state, X_n is the complete first-layer OU state, A_Z is the recurrent matrix, B_Z is the input matrix, b_Z is a bias vector, and ψ is a fixed feature map that may contain selected linear, quadratic and cross-bank features. The matrices and bias are drawn or designed once and then held fixed. A sufficient contraction condition for the echo-state property is

$$(1 - \alpha) + \alpha \|A_Z\|_2 < 1. \quad (7)$$

The echo-state property is important because it gives the recurrent layer fading memory: for the same input history, the effect of two different initial states decays geometrically. The state is therefore a stable causal function of the exogenous OU history rather than an additional source of persistent initial-condition dependence. This prevents exploding or ambiguous recurrent dynamics and makes simulations, calibration results and path reuse reproducible after a finite washout period.

Because Z is driven only by the exogenous OU state, neither X nor Z changes when a readout coefficient changes. The layer is called optional precisely because it affects variance only through the term $\delta^\top Z_t$ in the readout below. When the loading vector $\delta = 0$, that term vanishes and Layer 2 can be omitted without changing V or S . When $\delta \neq 0$, bounded nonlinear combinations of the OU history enter log variance and the full two-layer ESN is active. The recommended first calibration uses $\delta = 0$ and activates the layer only if an out-of-sample ablation demonstrates an incremental gain large enough to justify the extra parameters and the loss of Gaussian conditional-moment formulas.

Terminology

With $\delta = 0$, the operative model is a structured Gaussian reservoir rather than a nonlinear ESN in the narrow, classical sense. It remains an ESN-style fixed-reservoir model. The full two-layer ESN is obtained when $\delta^\top Z_t$ enters the readout.

2.4 Exponential-quadratic readout

The multiplier-free log-variance score is

$$\eta_t = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2 + \frac{q_O}{N_O} \|X_t^O\|^2 + \delta^\top Z_t. \quad (8)$$

The score η_t is dimensionless. The vector X_t^C contains the slow clustering coordinates and c_C is their linear loading. The non-negative curvatures q_Z , q_J and q_O control respectively the feedback, fast-spike and optional orthogonal quadratic channels. The constants β and β_J shift the vertices of the feedback and fast parabolas. The vector X_t^O contains N_O orthogonal coordinates, so division by N_O keeps their average squared energy comparable across bank sizes. Finally, δ is the loading vector of the optional echo-state layer Z_t .

The recommended core imposes $q_Z > 0$, $q_J \geq 0$, $\beta_J \geq 0$, and starts at $q_O = 0$, $\delta = 0$. Setting $\beta_J = 0$ centres the fast parabola at the stationary state $F = 0$. Its conversion of positive and negative fast-factor movements of equal magnitude into volatility spikes is then symmetric. A positive β_J , together with positive fast-factor/equity correlation, breaks this symmetry in the empirically relevant direction: negative index movements produce larger volatility spikes. Each shifted quadratic is deliberately parameterised without an additional linear loading on the same projection: adding both ℓF and $q_J(F - \beta_J)^2$, for example, would make the linear loading and shift redundant up to deterministic normalisation.

For a cap level L and sharpness $\kappa > 0$, define the smooth upper tempering function

$$h_{L,\kappa}(x) = L - \frac{1}{\kappa} \log\left(1 + e^{\kappa(L-x)}\right), \quad G_{L,\kappa}(x) = e^{2h_{L,\kappa}(x)}. \quad (9)$$

It satisfies $h_{L,\kappa}(x) \approx x$ well below L and $h_{L,\kappa}(x) \uparrow L$ as $x \rightarrow \infty$. Let $\xi_0(t)$ be the market forward variance curve and let $\underline{v}$ be a small deterministic variance floor. The variance is

$$V_t = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t)}{M_{\mathbb{Q}}(t)}, \quad M_{\mathbb{Q}}(t) = \mathbb{E}_0^{\mathbb{Q}}[G_{L,\kappa}(\eta_t)]. \quad (10)$$

Here L is the upper level approached by the tempered score, κ controls how sharply the transition occurs, $G_{L,\kappa}$ maps the score into a positive variance multiplier, and $M_{\mathbb{Q}}(t)$ is its time- t risk-neutral expectation conditional on time-zero information. The division by $M_{\mathbb{Q}}(t)$ preserves the prescribed forward variance curve.

Provided $\xi_0(t) > \underline{v}$, positivity and $\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t)$ follow by construction. The untempered shadow model is obtained by setting $h(x) = x$; it is useful for analytic checks and control variates, but it is not the recommended production specification on a return-correlated quadratic bank.

Tempering is introduced because an exponential of a Gaussian quadratic can assign enormous variance to rare factor realisations. Without tempering, admissibility depends on spectral moment conditions: some parameter values have infinite variance moments, Monte Carlo estimates can be dominated by a handful of paths, numerical optimisation becomes unstable, and the stock martingale condition is no longer automatic. The upper bound supplied by $h_{L,\kappa}$ ensures finite variance moments of every order and gives a direct finite-horizon Novikov bound, while leaving ordinary observations almost unchanged when $\eta_t \ll L$. What is sacrificed is exact Gaussian exponential-quadratic transform formulae and completely unbounded spike amplitudes. The untempered shadow model is therefore retained for analytic diagnostics, whereas the tempered model is used for robust simulation and calibration.

2.5 Equity dynamics under $\mathbb{P}$ and $\mathbb{Q}$

Let r_t be the instantaneous risk-free rate and d_t the continuous dividend yield. Under the physical measure, use a deterministic instantaneous Sharpe ratio $\theta(t)$:

$$\frac{dS_t}{S_t} = (r_t - d_t + \theta(t)\sqrt{V_t}) dt + \sqrt{V_t} dW_t^{S,\mathbb{P}}. \quad (11)$$

The name Sharpe ratio is literal at the instantaneous level: the expected equity return in excess of $r_t - d_t$ is $\theta(t)\sqrt{V_t}$, while its instantaneous volatility is $\sqrt{V_t}$. Their ratio is therefore $\theta(t)$. This specification lets the physical equity premium vary with volatility without feeding the variance level back into the exogenous reservoir dynamics.

Collect the independent primitive Brownian motions into a vector $\mathbf{W}$, whose entries are W^S, B^F, B^J, B^C and B^O , with as many components in the last two groups as their banks require. Let this vector satisfy the deterministic Girsanov shift

$$d\mathbf{W}_t^{\mathbb{P}} = d\mathbf{W}_t^{\mathbb{Q}} - \gamma(t) dt, \quad \gamma_S(t) = \theta(t). \quad (12)$$

The vector $\gamma(t)$ contains the market prices of risk of the independent Brownian sources. Its equity component γ_S must equal θ so that the physical equity premium disappears under $\mathbb{Q}$.

The other components need not equal θ : they price shocks that are not fully spanned by the equity return and determine how the conditional means of the volatility factors differ between $\mathbb{P}$ and $\mathbb{Q}$. Thus θ is both the instantaneous equity Sharpe ratio and one component of the broader Brownian risk-premium vector; the remaining γ 's refer to different primitive shock equations, not to alternative definitions of the equity Sharpe ratio.

Then

$$\frac{dS_t}{S_t} = (r_t - d_t) dt + \sqrt{V_t} dW_t^{S,\mathbb{Q}}. \quad (13)$$

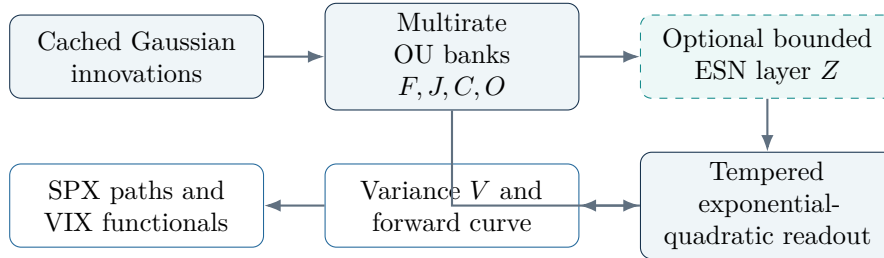
The unspanned components of γ are parsimonious variance-risk-premium parameters. Because γ is deterministic, every OU bank remains Gaussian under $\mathbb{Q}$: only its deterministic conditional mean changes, while its covariance and rate spectrum are preserved. This is the essential reason the same centred Gaussian innovations can be reused under both measures.

More explicitly, if $\gamma_b(t)$ denotes the effective deterministic risk premium of the Brownian driver assigned to coordinate (b, j) , then

$$dX_{j,t}^{b,\mathbb{Q}} = \left(-\lambda_j^b X_{j,t}^{b,\mathbb{Q}} - \sqrt{2\lambda_j^b} \gamma_b(t) \right) dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{Q}}. \quad (14)$$

For example, the feedback driver in the first equation is a linear combination of two primitive Brownian motions, so its effective premium is $\gamma_F = \rho_F \theta + \sqrt{1 - \rho_F^2} \gamma_{BF}$; the fast-bank formula is analogous. This is why the coordinate-level γ_b 's can differ even though the equity component is fixed by $\gamma_S = \theta$.

2.6 Architecture at a glance



3 Parameter hierarchy and path reuse

3.1 Hyperparameters, structural parameters and daily inputs

Class	Examples	Treatment
Architectural hyperparameters	Bank dimensions; OU rate grids; reservoir matrices A_Z, B_Z ; leak α ; feature map ψ ; cap family	Selected in an outer validation loop and changed infrequently. Changing OU rates or fixed reservoir weights requires rebuilding state features.

Class	Examples	Treatment
Shared structural parameters	$a, f, c_C, q_Z, \beta, q_J, q_O, \delta, \rho_F, \rho_J$	Estimated from a panel of physical and option data with regularisation. Readout changes do not change cached states; correlation changes require rebuilding correlated states from the same primitive innovations.
Measure-dependent parameters	θ and unspanned components of γ	Thin deterministic parameterisation. They shift conditional means but do not change OU covariances.
Daily inputs and states	$\xi_0(\cdot)$, rates, dividends, spot, filtered X_t, Z_t	Updated daily. They are inputs or filtered states, not structural calibration parameters.

3.2 What path reuse does and does not mean

For fixed hyperparameters and correlations, simulate or reconstruct once:

$$\{\Delta W^S, \Delta B^F, \Delta B^J, \Delta B^C, \Delta B^O; X; Z\}. \quad (15)$$

When $q_Z, \beta, q_J, q_O, c_C, \delta$ or the deterministic risk premia change, no new random numbers are required. The readout, variance, stock integral and payoffs must nevertheless be recomputed. It is impossible for the final S or VIX paths to remain unchanged when the volatility readout changes.

Change	New random draw?	Required recomputation
Readout coefficient	No	η, V, S , VIX map and payoffs
Deterministic risk premium	No	Deterministic OU mean shift; possibly Z ; then pricing quantities
Forward-variance curve	No	Variance level, S , VIX and payoffs
Correlation	No, if primitive shocks are cached	Correlated drivers and affected OU states
OU rate grid or reservoir matrix	No new seed is necessary, but the state changes	Full feature-engine reconstruction

This is the same computational principle exploited by precomputed signature and quantisation approaches: expensive stochastic input is generated offline, while parameter-dependent maps are evaluated during optimisation [4, 2].

4 Physical-measure stylised facts

The selected targets are standard empirical regularities of equity markets, with roughness and time-reversal asymmetry adding more discriminating requirements to the classical inventory [3, 5, 8].

4.1 Negligible linear autocorrelation of returns

For non-overlapping increments over a short interval Δ , the martingale part of equation (11) has conditional mean zero and variance of order Δ . The predictable return component is of order Δ , so its contribution to return covariance is of order Δ^2 . Consequently, linear return autocorrelation is small at intraday and daily frequencies unless the risk-premium term is made too large or too state dependent.

Under $\mathbb{Q}$, discounted simple gains are martingale increments and therefore orthogonal across non-overlapping intervals. Log returns need not be exactly uncorrelated because their drift contains $-V_t/2$.

Status: P/O. Near-zero autocorrelation is structurally natural, not an identity under $\mathbb{P}$. It must remain an explicit diagnostic, especially after estimating θ .

4.2 Volatility clustering

The recommended architecture obtains clustering from the exogenous slow bank rather than the endogenous activity filter. If $C_t = c_C^\top X_t^C$ and the coordinates are independent stationary OU processes, then

$$\text{Cov}(C_t, C_{t+\tau}) = \sum_{j=1}^{N_C} c_{C,j}^2 e^{-\lambda_j^C \tau}. \quad (16)$$

A geometrically spaced rate grid approximates slowly decaying covariance over a prescribed finite interval. The monotone exponential link transfers persistence from η to V . The quadratic readout enriches this set of persistence scales without introducing new state processes. For a centred stationary OU coordinate,

$$\text{Cov}\left((X_{j,t}^C)^2, (X_{j,t+\tau}^C)^2\right) = 2e^{-2\lambda_j^C \tau},$$

so a diagonal square contributes a mode with decay rate $2\lambda_j^C$. For two independent coordinates $i \neq j$, the cross-product channel satisfies

$$\text{Cov}\left(X_{i,t}^C X_{j,t}^C, X_{i,t+\tau}^C X_{j,t+\tau}^C\right) = e^{-(\lambda_i^C + \lambda_j^C)\tau}.$$

Thus $2\lambda_j^C$ is generated by the diagonal term $(X_j^C)^2$, whereas $\lambda_i^C + \lambda_j^C$ is generated by a cross term $X_i^C X_j^C$. These are two kinds of quadratic feature, not two additional channels of randomness. The diagonal features are already present in $q_C \|X^C\|^2$; cross terms arise automatically in a squared projection such as $(c^\top X^C)^2$, or can be introduced selectively through a symmetric quadratic matrix. They should be used only when the linear OU mixture cannot match the shape of the volatility autocorrelation function, because too many free cross terms impair identifiability.

This choice preserves strong frozen-state reuse. A conceptually different alternative is an endogenous activity bank

$$d\Sigma_{k,t} = \alpha_k(V_t - \Sigma_{k,t}) dt \quad (17)$$

where each $\Sigma_{k,t}$ is an exponentially weighted moving average of past variance and $\alpha_k > 0$ is its adjustment rate. This state is not part of the recommended specification in equation (8): equation (17) records a challenger model rather than an instruction to add Σ to the present readout. If the exogenous slow bank cannot fit the empirical volatility autocorrelation function (ACF), one may test the challenger by augmenting the score with parsimonious centred loadings such as $\sum_k b_{\Sigma,k}(\Sigma_{k,t} - \mathbb{E}[\Sigma_{k,t}])$. In discrete time, update $\Sigma_{k,n+1}$ from the already computed V_n , then use $\Sigma_{k,n+1}$ in the next variance readout. No additional ESN layer or random driver is needed. The cost is that these activity paths depend on the readout parameters and must be recomputed during calibration, so the strongest form of frozen-state reuse is lost.

Status: C/P/O. Persistence is built into the rate mixture; matching its empirical magnitude and shape is a calibration question.

4.3 Finite-scale roughness

Rough volatility is usually expressed through the small-lag scaling

$$\mathbb{E}[(\log V_{t+\tau} - \log V_t)^2] \asymp C\tau^{2H}, \quad H < \frac{1}{2}, \quad (18)$$

with empirical estimates often near $H = 0.1$ [5]. A fractional kernel can be approximated on a chosen range by a positive exponential sum,

$$K_H(t) \approx \sum_{j=1}^{N_F} a_j e^{-\lambda_j^F t}, \quad (19)$$

which is precisely a Markovian OU lift [1]. The feedback projection $Y = a^\top X^F$ therefore reproduces equation (18) over the frequencies covered by the rate grid.¹

To make the effect of the nonlinear variance readout precise, hold the other state coordinates fixed and regard log variance as a function of the rough feedback projection Y . In the tempered specification this one-dimensional map is

$$g_t(y) = \log \left(\underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t(y))}{M_{\mathbb{Q}}(t)} \right), \quad \eta_t(y) = \eta_t^{(-Y)} + q_Z(y - \beta)^2,$$

where $\eta_t^{(-Y)}$ collects every term in the score that does not vary with y . Thus g_t is not an additional model ingredient: it is simply the composition of the quadratic score, the smooth tempering function and the logarithm already present in equation (10). If g_t is twice continuously differentiable, Taylor's theorem gives

$$g_t(Y_{t+\tau}) - g_t(Y_t) = g'_t(Y_t)\Delta_\tau Y_t + \frac{1}{2}g''_t(\tilde{Y}_{t,\tau})(\Delta_\tau Y_t)^2, \quad (20)$$

$$\Delta_\tau Y_t := Y_{t+\tau} - Y_t,$$

¹Equation (19) approximates the fractional memory kernel by exponential modes. Each mode is realised by one coordinate of the OU bank X^F , and the same weights a_j combine those coordinates into $Y = a^\top X^F$. Consequently the increment variance of Y inherits the finite-band τ^{2H} scaling stated for log variance in equation (18); the smooth readout argument below explains why passing from Y to $\log V$ preserves that exponent locally.

for some intermediate point $\tilde{Y}_{t,\tau}$ between Y_t and $Y_{t+\tau}$. Over the scale range on which the OU lift approximates a rough kernel, $\Delta_\tau Y_t = O_{\mathbb{P}}(\tau^H)$. Provided g_t'' remains locally bounded, the second term in equation (20) is therefore $O_{\mathbb{P}}(\tau^{2H})$, which is $o_{\mathbb{P}}(\tau^H)$. Squaring the leading term and taking expectations consequently gives

$$\mathbb{E}[(\Delta_\tau \log V_t)^2] \approx \mathbb{E}[g_t'(Y_t)^2 (\Delta_\tau Y_t)^2] \propto \tau^{2H}.$$

The exponent being preserved is therefore the local roughness exponent: H for the size of log-variance increments, or equivalently $2H$ for their second moment in equation (18). The smooth transformation changes the multiplicative coefficient, not this power of τ , whenever $g_t'(Y_t) \neq 0$. The derivative can vanish at the quadratic vertex and becomes small near cap saturation, so roughness should be estimated on the bulk of the volatility distribution.

Finite lift, finite claim

Every finite OU system is a semimartingale and has Brownian $H = 1/2$ behaviour at sufficiently small asymptotic lags. The defensible claim is rough behaviour on the intraday-to-daily scale range selected by $[\lambda_{\min}^F, \lambda_{\max}^F]$, not genuine fractional asymptotics at arbitrarily small times.

Status: P/O. The lift supports the target scaling over a finite band; the realised band and estimated H must be verified with a noise-robust realised-volatility estimator.

4.4 Leverage effect

For a lag $\tau > 0$, define the leverage statistic

$$L(\tau) = \text{Corr}_{\mathbb{P}}(r_t, V_{t+\tau}), \quad L(\tau) < 0 \text{ empirically.} \quad (21)$$

The feedback contribution is $q_Z(Y - \beta)^2$, with derivative

$$\frac{\partial \eta}{\partial Y} = 2q_Z(Y - \beta). \quad (22)$$

Near the stationary centre $Y \approx 0$, its sign is $-\text{sign}(q_Z\beta)$. The approximation below follows directly from the diffusion terms. Because the coordinates entering $Y = a^\top X^F$ share W^F , its Brownian coefficient is $\nu_Y := \sum_j a_j \sqrt{2\lambda_j^F}$, so $dY_t = \text{drift } dt + \nu_Y dW_t^F$. The Brownian part of the equity log return is $\sqrt{V_t} dW_t^S$, with $d\langle W^S, W^F \rangle_t = \rho_F dt$. Applying Itô's formula to $\log V_t = g_t(Y_t)$, where g_t is exactly the same state-to-log-variance function defined in Section 4.3, and holding the other channels fixed, therefore yields

$$d\langle \log S, \log V \rangle_t = \sqrt{V_t} g_t'(Y_t) \nu_Y \rho_F dt.$$

Away from floor and cap saturation, the variance link is increasing in the score, hence

$$\text{sign}(g_t'(Y_t)) = \text{sign}(2q_Z(Y_t - \beta)).$$

If the positive rough-kernel weights make $\nu_Y > 0$ and the stationary state is concentrated near $Y = 0$, the leading sign is consequently

$$\text{sign}(d\langle \log S, \log V \rangle_t) \approx -\text{sign}(q_Z\beta\rho_F). \quad (23)$$

Under the present convention, $q_Z > 0$, $\beta > 0$, $\rho_F > 0$ therefore produce negative leverage. This apparently unusual positive correlation is a consequence of the shifted parabola convention; reversing both β and ρ_F leaves the local sign unchanged. The approximation neglects contributions from any other return-correlated variance channel, replaces the random factor $Y_t - \beta$ by its sign near the stationary centre, and assumes the tempering derivative has not become negligible. The exact simulated leverage statistic can therefore differ in magnitude and, if those qualifications fail, potentially in sign.

The shift is essential. With $\beta = 0$, the quadratic channel is locally even around the stationary centre and does not generate a stable directional leverage effect. The fast bank should not dominate total variance, because a large orthogonal or symmetric component dilutes correlation-based leverage statistics.

Status: P/N/O. The sign mechanism is explicit and has been observed in the programme's simulations. Its empirical magnitude remains a calibration constraint.

4.5 Strong Zumbach effect

The Zumbach effect is a causal time-reversal asymmetry between past returns and future volatility. A representative statistic at horizon τ is

$$Z(\tau) = \text{Corr}\left(R_{t-\tau,t}^2, \text{RV}_{t,t+\tau}\right) - \text{Corr}\left(\text{RV}_{t-\tau,t}, R_{t,t+\tau}^2\right). \quad (24)$$

Here $R_{a,b} := \log(S_b/S_a)$ is the cumulative log return over $[a, b]$. Realised variance (RV) over that interval is computed from a finer intraday partition $a = u_0 < \dots < u_m = b$ as

$$\text{RV}_{a,b} := \sum_{i=1}^m \left(\log \frac{S_{u_i}}{S_{u_{i-1}}} \right)^2.$$

In an ideal continuously observed diffusion, this sum converges as the mesh shrinks to the quadratic variation $\int_a^b V_s ds$. It is deliberately different from $R_{a,b}^2$: the latter squares the net movement and allows positive and negative intraday returns to cancel before squaring, whereas RV accumulates their individual squared movements. Unless stated otherwise, Corr in equation (24) is the ordinary Pearson correlation; the resulting $Z(\tau)$ is therefore the Pearson Zumbach asymmetry. A rank-correlation version is a distinct robustness diagnostic.

A reversible exogenous volatility process independent of returns has $Z(\tau) = 0$ in the ideal stationary limit. The current model breaks reversibility without making the OU driver amplitude depend on V :

1. The feedback state Y_t is a causal moving average of past innovations, including a component correlated with equity-return innovations. A return shock therefore remains in the state after the return interval has ended and can influence later volatility.
2. Future variance is a nonlinear, quadratic function of that persistent state. The square retains information about the magnitude of the earlier shock even when its sign is averaged out; the shift β additionally supplies directional leverage.
3. Consequently, a large past squared return predicts future RV through the state carried forward in time. Reversing the ordering asks whether past RV predicts a future squared return, but there is no reverse-time filter carrying future return shocks into the past. The two correlations in equation (24) need not coincide, and their difference is the measured causal asymmetry.

A new common-random-number experiment was run directly with the package implementation. Its results are:

Configuration	Mean $\pm$ SD	95% CI
Recommended model	0.180 ± 0.014	[0.170, 0.190]
Feedback quadratic removed ($q_Z = 0$)	0.255 ± 0.009	[0.248, 0.261]
Fast quadratic removed ($q_J = 0$)	0.029 ± 0.011	[0.021, 0.036]
Both correlated quadratics removed	-0.003 ± 0.006	[-0.007, 0.002]
All return-state correlations removed	-0.001 ± 0.015	[-0.012, 0.010]
GARCH(1,1) positive control	0.157 ± 0.008	[0.151, 0.163]

The table reports the Pearson statistic in equation (24). The ten fixed seeds were 710003, 730003, 750019, 770011, 790021, 810013, 830003, 850009, 870007 and 890023. The design used 32 paths per seed, ten simulated years, a two-year burn-in, 252 daily observations per year and two integration steps per day. For each path, $Z(\tau)$ was computed at $\tau = 5, 10, 20$ trading days using the model-integrated daily average variance as the noise-free limit of RV; the three horizons were averaged within each path, then the 32 paths were averaged within each seed. “SD” is the sample standard deviation across the ten seed means, and the confidence interval (CI) is the two-sided 95% Student- t interval across seeds. Every model ablation reused the same primitive Gaussian draws. The complete protocol, executable runner, per-seed values and summary are stored in `experiments/pearson_zumbach_001`.

The experiment confirms that return-state coupling and the correlated quadratic channels jointly create the asymmetry: both null specifications are statistically compatible with zero, whereas the GARCH control has the expected positive sign. It also corrects an overly simple interpretation of the mechanism. In the current parameterisation, removing only q_Z increases rather than removes Pearson asymmetry, while removing q_J reduces it sharply. Thus the fast shifted quadratic is the dominant contributor to this particular statistic, and the feedback and fast channels interact non-additively. This is synthetic mechanism evidence, not an empirical estimate of the market Zumbach effect. The construction differs from endogenous price-feedback models such as quadratic rough Heston, but targets the same time-reversal asymmetry [8, 6].

Measurement requirement

The two sides of equation (24) must measure volatility in the same way. For a daily horizon, for example, future and past RV should each be sums of squared intraday returns over the relevant day. Replacing RV by one squared daily return is not equivalent: intraday gains and losses first cancel inside the daily return, which makes the volatility proxy much noisier and can conceal the time-directional signal. This loss of statistical power occurred even for a simulated generalised autoregressive conditional heteroskedasticity (GARCH) positive control in which an asymmetry should be detected. A credible result must therefore state the intraday sampling frequency, apply the identical estimator to past and future windows, and show that the procedure returns approximately zero for a reversible null and a positive value for a positive control. Results should be replicated over declared independent seeds, with the uncertainty convention stated explicitly.

Status: N/O. The common-seed ablation supports the simulated mechanism and identifies the fast quadratic as its dominant contributor at the tested parameter point. Its market magnitude still requires a correctly measured intraday-data study.

4.6 Heavy-tailed returns at relevant frequencies

In the current diffusion model, a short return over an interval on which variance changes little is approximately $r \approx \sqrt{V\Delta}\varepsilon$, with $\varepsilon \sim \mathcal{N}(0, 1)$. Conditional on V , the return is therefore approximately Gaussian; unconditionally, random variance mixes Gaussian distributions of different widths and creates heavier tails. To isolate the contribution of the exponential-quadratic readout, consider a deliberately simplified one-factor shadow of the present model and write its volatility as

$$\sigma = e^{qY^2}, \quad Y \sim \mathcal{N}(0, 1), \quad q > 0, \quad (25)$$

one has, up to a slowly varying logarithmic factor,

$$\mathbb{P}(\sigma > x) = \mathbb{P}\left(Y^2 > \frac{\log x}{q}\right) \asymp x^{-1/(2q)}(\log x)^{-1/2}. \quad (26)$$

This σ is not the production volatility. It drops the forward curve, normaliser, variance floor, tempering, shifts and all but one Gaussian factor; the marginal calculation also treats Y simply as a standard normal variable. By contrast, the recommended model uses

$$\sigma_t = \sqrt{V_t} = \left[\underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t)}{M_{\mathbb{Q}}(t)} \right]^{1/2},$$

where η_t contains several linear and shifted-quadratic projections, some driven by Brownian motions correlated with the equity return. Setting $h(x) = x$, suppressing the floor and deterministic scale, and retaining only qY^2 gives $\sigma = e^{qY^2}$. The scalar example is therefore exactly the corresponding marginal channel of the untempered shadow model. It is useful here because it isolates the tail mechanism and its moment boundary in closed form; it is not used as the production volatility and does not imply that the tempered multifactor model has the same asymptotic tail law.

Since $V = \sigma^2 = e^{2qY^2}$, the displayed probability follows by solving $e^{qY^2} > x$ for Y^2 and applying the Gaussian tail approximation. It explains why the untempered exponential-quadratic channel can generate much heavier observable tails than a direct quadratic variance map.

In the recommended model the feedback and fast OU banks share Brownian components with the equity return, so the volatility state and return innovation are indeed dependent. Conditional on the entire variance path, the stochastic integral $\int \sqrt{V_t} dW_t^S$ is then not generally Gaussian: conditioning on an integrand built partly from the same Brownian motion also reveals information about the integrator. Consequently, the scalar marginal tail of σ cannot be converted directly into an exact tail index for finite-horizon returns. Dependence can also produce skewness and leverage and can change return-moment thresholds, so return tails must be assessed from the joint state–return dynamics rather than from a variance-mixture formula alone. For the recommended tempered model this does not create a moment-explosion problem: bounded variance on a finite horizon gives finite return moments and Gaussian-type ultimate tail bounds. The intended heavy tails are therefore effective finite-frequency tails over observable quantiles, not a genuine asymptotic power law.

For $X \sim \mathcal{N}(m, \Sigma)$ and a quadratic matrix Q , the untempered first variance moment requires

$$I - 4\Sigma^{1/2}Q\Sigma^{1/2} \succ 0. \quad (27)$$

To see why equation (27) is necessary, assume that Q is symmetric and begin with the Gaussian integral for the first untempered variance moment. Up to the finite normalising constant of the Gaussian density,

$$\begin{aligned}\mathbb{E}[e^{2X^\top QX}] &\propto \int_{\mathbb{R}^d} \exp\left[-\frac{1}{2}(x-m)^\top \Sigma^{-1}(x-m) + 2x^\top Qx\right] dx \\ &= e^{-\frac{1}{2}m^\top \Sigma^{-1}m} \int_{\mathbb{R}^d} \exp\left[-\frac{1}{2}x^\top (\Sigma^{-1} - 4Q)x + x^\top \Sigma^{-1}m\right] dx.\end{aligned}$$

Let $A := \Sigma^{-1} - 4Q$. If $A \succ 0$, completing the square gives

$$-\frac{1}{2}x^\top Ax + x^\top \Sigma^{-1}m = -\frac{1}{2}(x - A^{-1}\Sigma^{-1}m)^\top A(x - A^{-1}\Sigma^{-1}m) + \frac{1}{2}m^\top \Sigma^{-1}A^{-1}\Sigma^{-1}m,$$

and the remaining integral is a finite Gaussian integral proportional to $\det(A)^{-1/2}$. If A has a non-positive direction, the exponent fails to decay quadratically along that direction and generically grows, so the integral is not finite. Finally,

$$A = \Sigma^{-1/2} \left(I - 4\Sigma^{1/2}Q\Sigma^{1/2} \right) \Sigma^{-1/2},$$

so congruence preserves positive definiteness and gives exactly equation (27). In spectral terms, every eigenvalue of $\Sigma^{1/2}Q\Sigma^{1/2}$ must be below $1/4$. The factor four appears because the score is exponentiated twice to obtain variance rather than volatility. If the condition fails in any direction, the Gaussian density decays too slowly to offset the positive quadratic exponent, and even $\mathbb{E}[V]$ diverges. Higher variance moments impose progressively stronger spectral conditions. The tempered link removes asymptotic moment explosion and replaces true power tails by a broad but ultimately truncated mixture. This is consistent with the stated objective: realistic heavy-tail behaviour over observable intraday and daily quantiles is more important than an exact asymptotic law.

The optional orthogonal block can increase log-volatility dispersion² without changing the largest quadratic eigenvalue when its loading is spread over many independent directions and each new eigenvalue remains below the existing maximum. It should be activated only if the correlated feedback and fast banks cannot match return kurtosis and volatility quantiles without damaging leverage or Zumbach.

Status: P/N/O. Heavy finite-frequency tails are a strong capability of the readout. Their quantitative fit and stability across aggregation horizons remain empirical tests.

4.7 Taylor effect

The Taylor effect requires

$$\text{Corr}(|r_t|, |r_{t+k}|) > \text{Corr}(r_t^2, r_{t+k}^2) \quad (28)$$

over a relevant set of lags. It is not implied by leverage, roughness or clustering alone. For any stationary series U_t , write $\rho_U(k) := \text{Corr}(U_t, U_{t+k})$. Thus $\rho_{|r|}(k) = \text{Corr}(|r_t|, |r_{t+k}|)$ and $\rho_{r^2}(k) = \text{Corr}(r_t^2, r_{t+k}^2)$, the two quantities compared in equation (28).

²Here dispersion means the spread of the distribution of $\log \sigma_t = \frac{1}{2} \log V_t$ across times or simulated paths, commonly summarised by its variance or standard deviation. Greater dispersion means that conditional return scales vary more widely, strengthening the Gaussian-mixture tails; it does not mean dispersion across different traded assets.

The benchmark representation $r_t = e^{X_t \varepsilon_t}$ is used because it is the simplest stochastic-volatility mixture in which both correlations can be calculated in closed form. Here X is a zero-mean stationary Gaussian log-volatility with variance s^2 and $\rho_k := \text{Corr}(X_t, X_{t+k})$, while $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ is independent of the complete process X . It is not the exact untempered model developed in Section 2: there $\log \sigma_t$ is quadratic in Gaussian OU factors and is generally non-Gaussian. It is a tractable local benchmark with the same broad structure—a persistent random log-volatility multiplies an independent short-run innovation—and is used only to expose the role of log-volatility variability.

The formulae follow from Gaussian exponential moments. Independence gives

$$\begin{aligned} \mathbb{E}|r_t| &= \sqrt{2/\pi} e^{s^2/2}, & \mathbb{E}[|r_t||r_{t+k}|] &= (2/\pi) e^{s^2(1+\rho_k)}, \\ \mathbb{E}[r_t^2] &= e^{2s^2}, & \mathbb{E}[r_t^2 r_{t+k}^2] &= e^{4s^2(1+\rho_k)}, \end{aligned}$$

where $\mathbb{E}|\varepsilon| = \sqrt{2/\pi}$, $\mathbb{E}\varepsilon^2 = 1$, $\mathbb{E}\varepsilon^4 = 3$, and

$$\mathbb{E}[e^{aX_t + bX_{t+k}}] = \exp\left(\frac{1}{2}s^2(a^2 + b^2 + 2ab\rho_k)\right).$$

Using $\text{Corr}(U, V) = \text{Cov}(U, V) / \sqrt{\text{Var}(U) \text{Var}(V)}$, stationarity, and $\mathbb{E}|r_t|^2 = \mathbb{E}[r_t^2] = e^{2s^2}$ yields

$$\rho_{|r|}(k) = \frac{(2/\pi) (e^{s^2\rho_k} - 1)}{e^{s^2} - 2/\pi}, \quad (29)$$

$$\rho_{r^2}(k) = \frac{e^{4s^2\rho_k} - 1}{3e^{4s^2} - 1}. \quad (30)$$

Equations (29) and (30) are therefore exact within this Gaussian log-volatility benchmark, not approximations for the full exponential-quadratic reservoir model. In the full model the same two correlations are evaluated numerically on simulated or observed returns.

At small s^2 , the first correlation grows more slowly, so the Taylor inequality fails. At $\rho_k = 0.9$, the crossover in this benchmark is approximately $s^2 = 0.081$, or $\text{sd}(\log \sigma) \approx 0.285$. The threshold changes with persistence and with departures from the Gaussian log-volatility benchmark.

The exponential-quadratic and optional multi-directional blocks provide the required dispersion more naturally than a direct quadratic variance readout. Nevertheless, Taylor must remain a diagnostic or low-weight target rather than a theorem of the architecture.

Status: P/N/O. The model can cross the analytically identified dispersion threshold; the sign and lag range must be checked after every joint calibration.

4.8 Fast volatility, downside asymmetry and VIX spike relaxation

The variable F here is exactly the fast projection introduced in equation (5), namely $F = f^\top X^J$. Consider an event at time t that leaves its post-event value at $F_t = f_0$. Thus f_0 is the shocked state immediately after the event, not the model's initial state at calendar time zero. If one effective OU rate λ_J dominates and subsequent innovations are set to their conditional mean of zero, mean reversion gives

$$F_{t+s} \approx f_0 e^{-\lambda_J s}. \quad (31)$$

For the actual multi-rate projection the corresponding conditional mean is $\sum_j f_j X_{j,t}^J e^{-\lambda_j^J s}$; equation (31) is a one-rate summary of that mixture, useful when a dominant rate or an effective fitted rate describes the relaxation.

Define V_{t+s} by equation (10), evaluated along this shocked fast-factor path while keeping the forward curve, normaliser and all other state coordinates the same as in a reference path. Let $\bar{V}_{t+s}$ denote that reference variance with the fast projection held at its resting value $F = 0$. To obtain the next approximation, use the untempered link $h(x) = x$ and neglect the variance floor. The common deterministic factors then cancel in the ratio, leaving

$$\frac{V_{t+s}}{\bar{V}_{t+s}} \approx \exp\left(2\{q_J(F_{t+s} - \beta_J)^2 - q_J\beta_J^2\}\right) = \exp\left(2q_J[F_{t+s}^2 - 2\beta_J F_{t+s}]\right).$$

Substituting the conditional mean in equation (31) gives

$$\frac{V_{t+s}}{\bar{V}_{t+s}} \approx \exp\left(2q_J \left[f_0^2 e^{-2\lambda_J s} - 2\beta_J f_0 e^{-\lambda_J s}\right]\right), \quad (32)$$

Equation (32) is therefore exact for the deterministic one-rate decay in the untempered, zero-floor comparison and approximate for the recommended model because it suppresses future noise, replaces a multi-rate decay by one rate, and ignores tempering and the floor. The full recommended variance is always obtained directly from equation (10). For equal-magnitude shocks $f > 0$,

$$\log \frac{V(F = -f)}{V(F = +f)} = 8q_J\beta_J f. \quad (33)$$

Thus $q_J > 0$, $\beta_J > 0$, and positive fast-bank/equity correlation make a negative equity shock produce a larger contemporaneous volatility response than an equal positive shock. The result is exact below the tempering region; the smooth cap can attenuate it in the far tail. The zero-shift restriction is the symmetric nested control.

The corresponding time derivative is

$$\frac{d}{ds} \log\left(\frac{V_{t+s}}{\bar{V}_{t+s}}\right) = -4\lambda_J q_J \left[f_0^2 e^{-2\lambda_J s} - \beta_J f_0 e^{-\lambda_J s}\right]. \quad (34)$$

For the downside shocks of interest ($f_0 < 0$), this derivative is negative and its magnitude is largest near the event before slowing progressively. Several fast and medium OU rates create a sharp front followed by a persistent shoulder. This temporal shape comes from the interaction of mean reversion with the exponential level map, not from a jump in the state process.

Fresh-seed mechanism screen

A paired screen of 16 previously selected configuration–scenario pairs, five fresh seeds and six values $\beta_J \in \{0, 0.1, \dots, 0.5\}$ produced 480 evaluations. At $\beta_J = 0.1$, the downside-minus-upside median day-zero volatility-response gap increased from 35.0% to 38.7%, and the corresponding days 1–5 gap increased from 29.4% to 37.3%. All seven pre-existing stylised-fact proxy gates passed in 67 of 80 evaluations; six of 16 configurations passed on every fresh seed. All failures at this shift came from the positive-Zumbach lower bound. This is controlled synthetic mechanism evidence, not an empirical SPX calibration.

The cap may flatten the very top of an extreme spike. Its level should be selected so that it almost never binds in the calibration region while still controlling martingality outside it.

Status: P/N/O. The relaxation shape follows from the model and the downside orientation has controlled numerical support. Matching the distribution, amplitude and equity co-movement of observed VIX spikes is an empirical requirement.

4.9 Summary of physical-measure support

Target	Status	Load-bearing component
Near-zero return ACF	P/O	Diffusive return innovation and parsimonious deterministic Sharpe ratio
Volatility clustering	C/P/O	Slow exogenous OU mixture
Finite-scale roughness	P/O	Shared-driver rough OU lift and appropriate rate grid
Leverage	P/N/O	Shift β , curvature q_Z , and return correlation ρ_F
Strong Zumbach	N/O	Causal correlated OU filter plus quadratic readout
Heavy finite-frequency tails	P/N/O	Exponential quadratic; optional orthogonal dispersion block
Taylor effect	P/N/O	Sufficient log-volatility dispersion and persistence
Fast asymmetric VIX-like spikes	P/N/O	Shifted fast, partly correlated OU projection in the exponential quadratic

5 Risk-neutral pricing and joint SPX–VIX calibration

5.1 Forward variance, VIX and VIX futures

By equation (10), the input curve satisfies

$$\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t). \quad (35)$$

For a VIX window $\Delta = 30/365$, ignoring the conventional discrete-sampling and corridor adjustments,

$$\text{VIX}_T^2 = \frac{100^2}{\Delta} \int_T^{T+\Delta} \mathbb{E}_T^{\mathbb{Q}}[V_u] du. \quad (36)$$

Matching ξ_0 does not automatically match VIX futures, because

$$F_0^{\text{VIX}}(T) = \mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T] \leq \sqrt{\mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T^2]}. \quad (37)$$

The difference is the VIX convexity adjustment and must be calibrated or reported, not hidden inside the forward-variance normalisation.

5.2 Analytic Gaussian-core moments

If $\delta = 0$ and $h(x) = x$, the OU state is conditionally Gaussian and the score is quadratic. If

$$X_u \mid \mathcal{F}_T \sim \mathcal{N}(m_{T,u}, \Sigma_{T,u}) \quad (38)$$

and the exponent is $X^\top AX + b^\top X + c$, then

$$\mathbb{E}_T \left[e^{X^\top AX + b^\top X + c} \right] \quad (39)$$

$$= \frac{\exp \left(c + m^\top Am + b^\top m + \frac{1}{2} (2Am + b)^\top (I - 2\Sigma A)^{-1} \Sigma (2Am + b) \right)}{\sqrt{\det(I - 2\Sigma A)}}, \quad (40)$$

provided

$$I - 2\Sigma^{1/2} A \Sigma^{1/2} \succ 0. \quad (41)$$

This turns equation (36) into a one-dimensional deterministic maturity quadrature evaluated at the cached state X_T . No nested Monte Carlo is needed in the Gaussian shadow model.

5.3 Tempering and the nonlinear ESN layer

The recommended tempered model does not retain the closed form equation (40). Neither does the full model with $\delta \neq 0$. It remains computationally tractable through one of the following offline–online decompositions:

1. low-dimensional Gauss–Hermite or sparse-grid quadrature when the active quadratic directions are few;
2. functional quantisation of the Gaussian state;
3. least-squares conditional expectation on cached states;
4. a neural surrogate trained on the state and parameter domain, with the Gaussian core used as a control variate.

Quantisation and neural pricing have already been demonstrated for Gaussian polynomial SPX–VIX models [2]; precomputed signature samples provide another successful offline calibration paradigm [4]. These are useful numerical templates, not proofs that the present model will calibrate.

5.4 SPX pricing and martingality

For fixed cached innovations and a candidate parameter vector, compute

$$\log \frac{S_T}{S_0} = \int_0^T \left(r_t - d_t - \frac{1}{2} V_t \right) dt + \int_0^T \sqrt{V_t} dW_t^{S, \mathbb{Q}}. \quad (42)$$

The discounted stock is always a nonnegative local martingale. It is a true martingale if the stochastic exponential has expectation one. In the tempered model, V is bounded on every finite horizon when ξ_0 and the deterministic risk premia are bounded and the positive normaliser $M_{\mathbb{Q}}$ is bounded away from zero. Novikov’s condition then gives

$$\mathbb{E}^{\mathbb{Q}} \left[\exp \left(\frac{1}{2} \int_0^T V_t dt \right) \right] < \infty, \quad (43)$$

so the discounted stock is a true martingale. In the untempered, return-correlated exponential-quadratic model, finite marginal moments do not by themselves establish true martingality; the admissible domain must be analysed separately.

Production choice

The smooth upper tempering is part of the recommended model, not merely a numerical emergency stop. It provides a clean finite-horizon martingale argument while preserving the untempered map over the empirically relevant state region.

5.5 Why a quintic polynomial is not structurally required

The fifth-degree polynomial used by Abi Jaber, Illand and Li is effective in their one-factor Gaussian polynomial family and helps generate upward VIX smiles [2]. It is not a necessity theorem. Two complementary results make this clear:

- quadratic rough Heston combines a quadratic variance map with rough feedback and provides a continuous-path solution to the joint SPX–VIX smile problem [6];
- signature models obtain strong joint calibrations using several low-order path features rather than a scalar quintic readout [4].

The present architecture substitutes feature diversity, rate diversity and exponential convexity for scalar polynomial degree. Its fast and slow quadratic directions have distinct economic roles, which should improve identifiability relative to an unrestricted high-degree polynomial.

6 Joint calibration programme

6.1 Recommended nested sequence

Model	Readout	Purpose
M0	Positive direct quadratic variance	Minimal baseline; tests whether exponential convexity is genuinely needed.
M1	Untempered Gaussian exponential quadratic, $\delta = 0$	Analytic shadow model; moment-domain and VIX-control-variate benchmark.
M2	Tempered Gaussian exponential quadratic, $\delta = 0$	Recommended production core.
M3	M2 plus $\delta^\top Z$	Activate only if the nonlinear ESN layer materially improves out-of-sample physical facts or option errors.
M4	M3 plus orthogonal block $q_O > 0$	Optional finite-frequency dispersion and tail tuning; activate only after diagnosing a residual deficiency.

If M2 cannot reproduce synthetic surfaces generated by a quintic OU benchmark, add a second identified quadratic direction before increasing polynomial degree. This directly tests the hypothesis that factor diversity can replace degree.

6.2 Relation of M2 to ESN–A2

M2 is a structured descendant of ESN–A2, but it is neither a renaming nor a nested restriction of A2. The common core is a fixed multirate Gaussian reservoir whose state can be reused while the volatility readout changes. The redesign concerns the organisation of that state, the readout parameterisation, and the separation between physical and pricing inputs. The comparison

below refers to the supplied reconstructed A2 engine; its implementation file records that it was reconstructed after an environment reset and validated against the reported roughness targets. Consequently, code-specific A2 conclusions carry that provenance.

On a grid of width Δ , the A2 rough bank and nonlinear layer can be written schematically as

$$r_{n+1} = D_\lambda r_n + c_\lambda \varepsilon_{n+1}, \quad D_{\lambda,jj} = e^{-\lambda_j \Delta}, \quad c_{\lambda,j} = \sqrt{1 - e^{-2\lambda_j \Delta}}, \quad (44)$$

$$z_{n+1} = \Phi_{A2}(r_{n+1}, z_n), \quad (45)$$

where the same scalar innovation ε_{n+1} drives every coordinate of r and the equity return in the supplied simulator. The map Φ_{A2} is the leaky, bounded second layer built from quadratic and linear features, a hyperbolic tangent, local recurrence, and the state-dependent norm gate. Its volatility score and variance map are

$$\eta_n^{A2} = b_0 + \kappa q^\top r_n + w_z^\top z_n + \frac{m_1}{N_r} \|r_n\|^2 + \frac{m_2}{N_r} (q^\top r_n)^2, \quad (46)$$

$$V_n^{A2} = \sigma_{\min}^2 + \text{softplus}(\eta_n^{A2})^2. \quad (47)$$

At the A2 resting bias, softplus is already close to its exponential branch. Thus A2 already contains the central idea “Gaussian reservoir plus approximately exponential-quadratic readout.” In contrast, M2 sets $\delta = q_O = 0$ in equation (8) and uses

$$\eta_t^{M2} = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2, \quad (48)$$

$$V_t^{M2} = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t^{M2})}{M_Q(t)}. \quad (49)$$

The models therefore share a mathematical backbone, but they are not the same parameterised family:

Component	ESN–A2	M2
State drivers	One multirate rough bank driven by the equity innovation; the coupling is effectively perfect in the supplied engine.	Separate feedback, fast-spike, and clustering banks. Feedback and spike shocks may be only partially correlated with the equity shock; clustering shocks are independent.
Nonlinear layer	The bounded recurrent z -layer contributes directly through $w_z^\top z$.	Inactive: $\delta = 0$. M3, not M2, restores an optional fixed contractive echo layer.
Leverage and Zumbach	The linear rough projection, aligned quadratic projection, and nonlinear layer have overlapping effects, so attribution is parameter-regime dependent.	Concentrated in the shifted correlated channel $q_Z(Y - \beta)^2$, whose slope controls the leverage sign and whose state dependence generates time-reversal asymmetry.
Clustering	Encoded jointly by the multirate bank, especially its isotropic reservoir-energy term, and persistence transmitted through the nonlinear layer.	A dedicated slow linear bank $c_C^\top X^C$, separated from the leverage–Zumbach channel.

Component	ESN–A2	M2
Fast VIX-like spikes	No separately identified fast-spike bank.	A dedicated convex direction $q_J(F - \beta_J)^2$, with its own fast rate grid and return correlation; $\beta_J > 0$ orients equal-magnitude responses toward downside shocks.
Variance map and tails	Squared softplus with an unbounded exponential branch; the quadratic eigenvalues govern its asymptotic moment domain.	A direct exponential-quadratic shadow map, smoothly tempered in production and close to the shadow map below the cap. The target is finite-frequency fat tails, not an asymptotic power law.
Measure change	The supplied engine is a physical-measure simulator and does not give a complete, separately normalised $\mathbb{P}/\mathbb{Q}$ specification.	Explicit $\mathbb{P}$ and $\mathbb{Q}$ stock equations, deterministic Girsanov kernel, unspanned risk premia, and a market forward-variance normalisation.
Path reuse	Available while parameters that determine r and z are fixed; changing reservoir or second-layer parameters rebuilds the state.	Primitive shocks and exogenous states are cached. Readout, forward-curve, and deterministic risk-premium changes require deterministic state adjustments and revaluation, but no new random draw.
Calibration roles	The audited parameter split allowed quantities intended for daily recalibration to move most of the physical statistics.	Architecture, shared structural parameters, measure-dependent premia, and daily inputs are separated explicitly; shared readout coefficients cannot be overwritten by a daily $\mathbb{Q}$ fit.

The shifted M2 channel makes the relationship particularly transparent:

$$q_Z(Y - \beta)^2 = q_Z Y^2 - 2q_Z \beta Y + q_Z \beta^2. \quad (50)$$

It combines A2’s aligned linear and quadratic rough-factor channels, but imposes the identified coefficient relation $\kappa_Y = -2q_Z \beta$ after normalisation. Conversely, baseline M2 drops A2’s isotropic energy and nonlinear readouts, replaces their clustering role by X^C , and adds the fast direction F . M4 can restore an independent quadratic dispersion channel, but that channel lives in an orthogonal bank and is therefore not equivalent to A2’s $\|r\|^2$. M3 restores nonlinear memory, but its controlled echo layer does not make the resulting model identical to A2 either.

Nesting statement

A2 is not a parameter point of M2, and M2 is not a parameter point of A2. M2 is best viewed as a lower-nonlinearity but more economically structured redesign of A2. It retains the exogenous Gaussian and exponential-quadratic mechanisms, while changing the driver topology, assigning separate channels to clustering and spikes, enforcing path-reuse discipline, and adding the objects required for joint SPX–VIX pricing.

The transfer of stylised-fact evidence is therefore conditional. A2 experiments motivate the

rough-reservoir and convex-readout mechanisms, but they do not prove that M2 preserves their empirical magnitudes. M2 must be evaluated on its own with matched estimators, multiple seeds, and the ablations prescribed below. Likewise, M2 has better specified joint-pricing capacity, not a demonstrated SPX–VIX fit in the absence of option data.

6.3 Calibration hierarchy

Let Θ_H , Θ_S , Θ_M and D_t denote architectural hyperparameters, shared structural parameters, measure-dependent parameters and daily inputs. A practical hierarchy is:

1. **Architecture selection.** Choose dimensions and rate grids using rolling physical-measure validation and synthetic option-capacity tests. This is the expensive outer loop.
2. **Shared structural estimation.** Estimate Θ_S from historical windows and a panel of option surfaces, with stability penalties across dates.
3. **Measure change.** Estimate the thin deterministic unspanned risk premia in Θ_M , with θ pinned by the physical drift convention.
4. **Daily calibration.** Strip ξ_0 , filter the current state and calibrate only the permitted daily pricing quantities.

Because shared readout parameters affect both $\mathbb{P}$ and $\mathbb{Q}$, a purely sequential "fit $\mathbb{P}$, then overwrite under $\mathbb{Q}$ " procedure is not valid. Use either a joint loss or alternating minimisation with a physical-measure retention penalty:

$$\mathcal{L}(\Theta) = w_P \mathcal{L}_P + w_S \mathcal{L}_{\text{SPX}} + w_V \mathcal{L}_{\text{VIX}} + w_F \mathcal{L}_{\text{VIX futures}} + \mathcal{R}(\Theta). \quad (51)$$

Option residuals should be expressed in bid–ask-normalised prices, for example

$$\varepsilon_i = \frac{C_i^{\text{model}} - C_i^{\text{mid}}}{\max\{\frac{1}{2}(C_i^{\text{ask}} - C_i^{\text{bid}}), \epsilon_i^{\text{floor}}\}}, \quad (52)$$

rather than unweighted implied-volatility errors. VIX futures must be included separately from the forward-variance curve.

6.4 Validation and falsification tests

Question	Required test	Failure implication
Roughness	Structure functions on noise-robust intraday realised volatility across the designed lag band	Retune rate grid; do not add readout degree.
Leverage	Lagged return–realised-volatility correlation with confidence bands	Adjust β , ρ_F , or add a separate linear leverage bank.
Zumbach	Realised-variance statistic, reversible null, positive control, multiple seeds or blocks	Reassess the correlated quadratic mechanism or measurement protocol.

Question	Required test	Failure implication
Heavy tails	QQ/body fit, kurtosis, threshold-stability plots and aggregation horizons	Add orthogonal dispersion only after excluding measurement and cap effects.
Taylor	Both correlation curves and the log-volatility dispersion threshold	Treat as secondary unless failure contaminates more important targets.
VIX spikes	Peak amplitude, rise/decay asymmetry and SPX co-movement	Retune fast rates and ρ_J ; avoid a dominant independent spike factor.
Martingale	Analytic boundedness proof plus Monte Carlo defect within sampling error	Reject parameter point or lower the cap.
Joint surface capacity	Bid-ask-normalised SPX/VIX errors and VIX-futures errors over dates	Add a second quadratic direction before quintic degree.
Path reuse	Bitwise equality of cached X, Z under readout perturbations	Implementation violates the architectural constraint.

7 Open issues and limitations

What is not yet established

1. No SPX/VIX option data have yet been used, so joint calibration capacity is untested.
2. Identifiability of curvature, shift, correlations, fast rates and deterministic risk premia must be studied through Jacobian singular values, profile likelihoods and rolling stability.
3. The optimal cap level must balance negligible in-sample binding against robust moment and martingale control.
4. A finite OU lift produces only finite-scale roughness.
5. The exact market magnitude of the Zumbach effect requires intraday realised variance; daily squared returns are an inadequate proxy for the current statistic.
6. The nonlinear second layer is not currently justified by the exploratory Zumbach ablation. It must earn its parameters through a reproducible out-of-sample improvement.

Three interactions deserve particular monitoring:

- Increasing an independent dispersion block can improve tails and Taylor while diluting leverage and Zumbach correlations.
- Increasing the correlated quadratic curvature strengthens Zumbach and tails but pushes the untempered model toward moment and martingale boundaries.
- A low cap improves safety but can flatten extreme VIX spikes, distort roughness estimates in the upper tail and limit the VIX smile.

These are not arguments against the model. They are the principal trade-offs that the joint objective must expose rather than hide.

8 Conclusion

The current candidate is a two-layer exogenous reservoir architecture with a recommended Gaussian first calibration and an optional nonlinear ESN extension. Its defining choices are:

- multirate OU banks with variance-independent drivers;
- a return-correlated rough feedback projection;
- a shifted quadratic feedback term for leverage and Zumbach;
- slow exogenous modes for clustering and path reuse;
- a shifted fast, partly return-correlated direction for downside-oriented VIX spike relaxation, nested at $\beta_J = 0$;
- a normalised, smoothly tempered exponential-quadratic variance map;
- a deterministic market price of risk preserving the Gaussian OU class under $\mathbb{Q}$;
- an optional independent dispersion block and optional nonlinear ESN layer, both subject to ablation.

This construction retains the mechanisms needed for the targeted physical-measure facts and offers a viable offline–online route to SPX/VIX calibration without a quintic polynomial. Roughness and path reuse are architectural; leverage has an explicit sign condition; Zumbach has controlled numerical support; tails and VIX spikes follow from exponential-quadratic convexity; and Taylor follows only above a measurable dispersion threshold.

The next decisive step is not another architectural elaboration. It is a disciplined implementation of M0–M2, validation on intraday physical data, and synthetic joint-calibration tests against quadratic-rough-Heston and quintic-OU benchmarks before real option data become available.

A Derivation of the Gaussian quadratic transform

Let $X = m + Z$, with $Z \sim \mathcal{N}(0, \Sigma)$, and let $A = A^\top$. Write

$$X^\top AX + b^\top X + c = c + m^\top Am + b^\top m + u^\top Z + Z^\top AZ, \quad u = 2Am + b. \quad (53)$$

Whitening with $Z = \Sigma^{1/2}U$ and completing the square in the Gaussian density gives

$$\mathbb{E}\left[e^{u^\top Z + Z^\top AZ}\right] = \det(I - 2\Sigma A)^{-1/2} \exp\left[\frac{1}{2}u^\top (I - 2\Sigma A)^{-1}\Sigma u\right], \quad (54)$$

provided $I - 2\Sigma^{1/2}A\Sigma^{1/2} \succ 0$. Substitution yields equation (40). For $e^{2\eta}$ with quadratic matrix Q , set $A = 2Q$, which gives the condition in equation (27).

B Derivation of the Taylor benchmark

Let $r_t = e^{X_t} \varepsilon_t$, where $\mathbb{E}[X_t] = \mu$, $\text{Var}(X_t) = s^2$, $\text{Corr}(X_t, X_{t+k}) = \rho_k$, and the standard-normal innovations are independent through time and independent of X . Then

$$\mathbb{E}[e^{X_t}] = e^{\mu+s^2/2}, \quad \mathbb{E}[e^{2X_t}] = e^{2\mu+2s^2}, \quad (55)$$

$$\mathbb{E}[e^{X_t+X_{t+k}}] = e^{2\mu+s^2(1+\rho_k)}, \quad \mathbb{E}[\varepsilon] = \sqrt{2/\pi}. \quad (56)$$

Using these moments in the covariance and variance of $|r|$ yields equation (29). Applying the same calculation to r^2 , with $\mathbb{E}[\varepsilon^4] = 3$, gives equation (30). The result is exact for this benchmark and serves as a diagnostic approximation for the current nonlinear model.

C Related model classes

Class	Main device	Relation to the present model
Gaussian polynomial volatility	Polynomial of a Gaussian Volterra state	Demonstrates strong joint SPX–VIX calibration and motivates Gaussian path reuse; degree five is specific to that one-factor family.
Quadratic rough Heston	Endogenous rough price feedback and quadratic variance	Shows that a quadratic map can support strong Zumbach and joint smiles, but sacrifices exogenous frozen states.
Signature volatility	Linear readout of rich path features	Demonstrates that feature diversity and offline samples can replace high scalar polynomial degree.
Rough-volatility lift	Exponential-sum approximation of fractional kernels	Supplies the multirate Markovian mechanism used for finite-scale roughness.
Classical ESN	Fixed contractive recurrent reservoir plus trained readout	Supplies the optional nonlinear second layer and the path-reuse philosophy.

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